SingleCellMultiModal: Integrating Multi-modal Single Cell Experiment datasets

https://github.com/waldronlab/SingleCellMultiModal

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Introduction

We present a new ExperimentHub package, *SingleCellMultiModal,* that provides ready-to-use datasets for benchmarking and methods development for single-cell multimodal technologies including CITE-seq, scNMT, and seqFISH.

Technology	Modalities measured
G&T-seq	Genome + mRNA
CITE-seq	Cell surface marker + mRNA (1)
ECCITE-seq	Cell surface marker + mRNA + CRISPR perturbations (2)
10X Multiome ATAC-seq	Chromatin accessibility + mRNA
scNMT	Epigenetic markers (chromatin accessibility, methylation) + mRNA (3)
seqFISH	mRNA in spatial context, *in situ* (4, 5)

Methods

SingleCellMultiModal provides pre-processed datasets as *MultiAssayExperiment* objects for coordinated management of multiple experimental assays performed on overlapping sets of cells and samples.

SCMM represents an array single cell methods

CITE-seq

Dataset Identifier	Assay Type	Modes	# features	# cells
Cord blood	RNA-seq	Transcripts	36280	8617
	Antibody cell surface tagging	Proteins	13	8617

Table 1
Description of available CITE-seq dataset in *SCMM,* data size 1.2Gb; matrix data structure

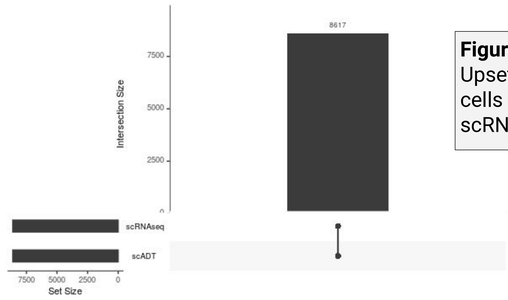

Figure 1
Upset plot of dataset in SCMM. All cells are matched in both modalities: scRNAseq and scADT.

scNMT

Dataset Identifier	Assay type	Modes	# features	# cells
scNMT	RNA-seq	Transcripts	18345	2480
	DNA Methylation	CpG islands	14080	986
		promoters	17179	986
		Gene bodies	17559	986
		DHS	6673	986
	Chromatin accessibility	CpG islands	14824	1101
		promoters	18037	1103
		Gene bodies	17924	1105
		DHS	20082	1094

Table 2
Description of available scNMT dataset in *SCMM,* data size 277.3Mb, matrix data structure

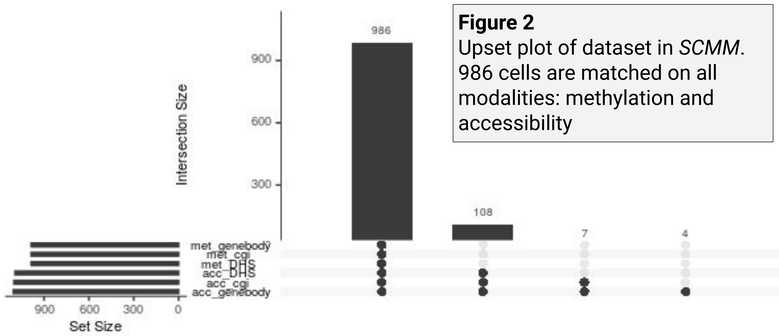

Figure 2
Upset plot of dataset in *SCMM.* 986 cells are matched on all modalities: methylation and accessibility

seqFISH

Dataset Identifier	Assay Type	Modes	Data Structure	Version	# features	# cells
Mouse visual cortex	scRNA-seq	Transcripts	SingleCellExperiment	1.0.0	24057	1809
				2.0.0	113	1723
	seqFISH	Spatial Transcripts	SpatialExperiment	1.0.0/ 2.0.0	113	1597

Table 3
Description of available seqFISH dataset in *SCMM,* data size 4Mb.

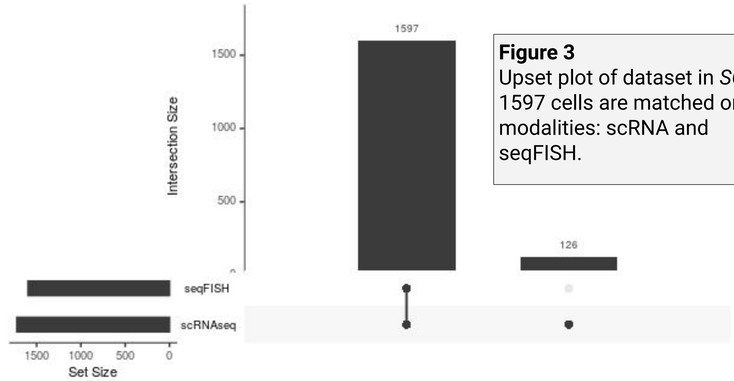

Figure 3
Upset plot of dataset in *SCMM.* 1597 cells are matched on all modalities: scRNA and seqFISH.

Conclusion

SingleCellMultiModal provides pre-integrated, ready-to-use benchmark datasets for developing and testing methods of single-cell multimodal data analysis for Bioconductor.

Next steps: Include ECCITE-seq, 10X Multiome ATAC-seq, and G&T-seq data sets in SCMM package

References

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